

# IMPLEMENTATION REVIEW

Batch No : B20

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Under the Guidance of :-  
Mrs. G. Manasa

# Tools & Technologies Used



- Python 3.7



- Flask (Backend & Routing)



- XGBoost (ML Classifier)



- HTML, Jinja2 Templates (Frontend)



- NumPy, Pandas, Scikit-learn, Joblib

# ML Model - XGBoost Classifier

- **Chosen for speed, tabular data performance, and accuracy:**  
XGBoost was selected due to its high efficiency and superior accuracy in handling structured datasets like ASD screening data.
- **Trained on ASD dataset (publicly available):** The model was trained using a publicly available Autism Spectrum Disorder dataset to ensure transparency and reproducibility.
- **Preprocessing:** Label encoding, standard scaling: Categorical variables were converted using label encoding, and numerical features were scaled to standardize the input space.

# ML Model - XGBoost Classifier

- **Cross-validated & fine-tuned:** The model underwent **k-fold cross-validation** and hyperparameter tuning to optimize performance and reduce overfitting.
- **Saved with Joblib for deployment:** The trained model was serialized using **Joblib** to enable seamless integration into the web application for real-time predictions.

# Data Preprocessing

- Identified and handled **missing** or **inconsistent** values in the dataset.
- Applied **Label Encoding** to convert categorical features (e.g., gender, jaundice, family history) into numerical form.
- Used **StandardScaler** to normalize numerical features and ensure consistent input ranges.
- Ensured the dataset was clean, structured, and ready for **training** the **XGBoost model**.
- Improved model accuracy and convergence by **standardizing** the input data.

# Frontend & UI

- Developed a **user-friendly web interface** to collect responses from parents or caregivers.
- Built using **HTML, CSS, and JavaScript** for responsive and accessible design.
- Integrated with **Flask backend** to send input data to the ML model for prediction.
- Designed **intuitive forms** with clear labels to ensure ease of use and reduce user error.
- Displayed prediction results along with **risk level** and precautionary **advice** in a clean layout.

# Backend Logic

- Implemented using **Flask** to handle web requests and connect frontend with the ML model.
- Receives **user input** from the frontend and processes it in **real time**.
- Applies the same **preprocessing steps** (label encoding, scaling) used during model training.
- Loads the **XGBoost model** saved with **Joblib** to generate predictions.
- Sends the predicted **autism risk level** back to the frontend along with suitable advice.
- Ensures smooth, secure, and efficient communication between **UI and ML** components.

# Flow of Execution

- **User accesses** the web application through a browser interface.
- **Fills out** the autism screening form with relevant information.
- Form data is **submitted to the Flask** backend via HTTP request.
- Backend **preprocesses** the **input** using label encoding and standard scaling.
- The cleaned data is fed into the **XGBoost model** to make a prediction.
- The predicted **risk level** and **advice** are returned to the frontend.
- Results are displayed to the user in an **easy-to-understand** format.



# Welcome Page

Autism Detection

[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

## Autism Detection in Toddlers

Early detection helps in timely interventions that enhance development.

[Start The Survey](#)

# Toddler Data Input Screen

**Autism Detection**[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

## Child Screening Form

What is your child's age (in months)?

What is your child's sex?

☐ Male ☐ Female

What is your child's ethnicity?

Select one

Was your child born with jaundice?

☐ Yes ☐ No

Does your child have an immediate family member with Autism Spectrum Disorder (ASD)?

☐ Yes ☐ No

Who is completing this screening test?

Select one

**Submit**

# Questionnaire Screen

Autism Detection

[Home](#) [Features](#) [Contact](#) [FAQs](#) [About](#)

## Question 1

Does your child look at you when you call his/her name?

Observe how your child responds when you call their name.

☐ Yes ☐ No

Next

# Predicted Results Page

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## Autism Screening Results

✓ Screening Result: **No**

Q-Chat Score

2

Risk Level

Low

### Identified Risk Questions

Q1

Does your child look at you when you call his/her name?

Q2

How easy is it for you to get eye contact with your child?

# Risk level display with brief recommendations

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## Recommended Treatment Options

 **Low Risk Treatment Options** 

**Parent Education and Training**

Learning strategies to support your child's development, communication, and behavior management at home.

[Learn more](#) 

**Developmental Monitoring**

Regular check-ups with pediatrician to track developmental milestones and address any concerns early.

[Learn more](#) 

**Speech and Language Therapy**

Helps develop communication skills, language comprehension, and social use of language.

[Learn more](#) 

**Social Engagement Activities**

Playgroups, storytime sessions, and other activities that promote social interaction and engagement.

[Learn more](#) 

 **Medium Risk Treatment Options** 

 **High Risk Treatment Options** 